

A codimension-one counterexample to uniform ball proximality

Autonomous proof-discovery packet for arXiv:1610.03209

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Status. Candidate full counterexample, likely valid; human review recommended. Model: GPT5.6.

1 Source question

For a closed convex subset C of a Banach space X , Paul uses the following definition: C is *uniformly proximal* if, for every $\varepsilon, R > 0$, there is $\delta(\varepsilon, R) > 0$ such that whenever

$$\text{dist}(x, C) \leq R, \quad y \in C, \quad \|x - y\| < R + \delta,$$

there is $y' \in C$ satisfying

$$\|y - y'\| < \varepsilon, \quad \|x - y'\| \leq R.$$

Problem 4.7 of [1] asks:

Let Y be a subspace of X which is uniformly proximal. Is it necessary that B_Y is also uniformly proximal in X ?

The source passage (PDF page 14) is reproduced below.

$\|y_n\| \leq \|x\| + \varepsilon + \delta(\varepsilon, R) = \|x\| + \varepsilon + 2\delta(\varepsilon, R) < R + \delta(\varepsilon, R)$
This completes the proof. $\square$

We now propose the following problem which is relevant to the subsequent matter.

Problem 4.7. *Let Y be a subspace of X which is uniformly proximal. Is it necessary that B_Y is also uniformly proximal in X ?*

Remark 4.8. (a) *It is clear from the Definition 4.2 that uniform proximality of C is a uniform version of strong proximality for the points which are of finite distance away from C . Hence due to the Example by Godefroy in [11] it is clear that closed unit ball of a Banach space not necessarily uniformly proximal.*

The immediately following Remark 4.8(a) recalls that some Banach-space unit balls are not uniformly proximal. Thus, under the convention that a subspace may equal its ambient space, $Y = X$ already gives a literal negative answer. The result below removes that wording issue completely: Y is proper and has codimension one, and the bad unit ball is given by an explicit elementary norm.

2 Proof intuition

There are two independent pieces. First, construct a normed space E whose unit ball has an extreme point u that is not strongly extreme. Unit vectors a_n can then remain far from u while both endpoints of the approximate chord through u stay asymptotically in the unit ball. Seen from $2u$, these a_n are approximate nearest points, whereas extremality makes u the unique exact nearest point. This is precisely the obstruction to uniform proximality.

Second, place E as the coordinate hyperplane in $E \oplus_\infty \mathbb{R}$. Coordinate hyperplanes in this geometry have the strongest elementary proximality modulus: move radially in the E coordinate until the distance is exactly R . At the same time, restricting to the zero scalar coordinate preserves every distance involved in the bad unit-ball configuration.

3 An explicit unit ball which is not uniformly proximal

Let $H = \ell_2(\{2, 3, \dots\})$ with canonical vectors $(e_n)_{n \geq 2}$, and put $E = \mathbb{R} \times H$. Define

$$\mathcal{N}(t, x) = \max \left\{ |t| + \sup_{n \geq 2} \frac{|x_n|}{n}, \|x\|_2 \right\}. \quad (1)$$

Lemma 1. *Equation (1) defines a Banach norm on E , and its closed unit ball B_E is not uniformly proximal in E .*

Proof. Both entries in the maximum are subadditive and absolutely homogeneous, and their common kernel is zero. Moreover

$$\max\{|t|, \|x\|_2\} \leq \mathcal{N}(t, x) \leq |t| + \|x\|_2.$$

Thus $\mathcal{N}$ is an equivalent Banach norm on $\mathbb{R} \times \ell_2$.

Write $u = (1, 0)$. We first show that u is an extreme point of B_E . Suppose

$$u = \frac{(s, x) + (2 - s, -x)}{2}, \quad (s, x), (2 - s, -x) \in B_E,$$

and set $\alpha = \sup_{n \geq 2} |x_n|/n$. The first entry of (1) gives

$$|s| + \alpha \leq 1, \quad |2 - s| + \alpha \leq 1.$$

Since $|s| + |2 - s| \geq 2$, adding the inequalities forces $\alpha = 0$. Hence $x = 0$, and $|s| \leq 1$, $|2 - s| \leq 1$ force $s = 1$. Therefore both endpoints equal u , proving extremality.

For $n \geq 2$, define

$$a_n = \left(1 - \frac{1}{n}, e_n\right).$$

Direct substitution gives

$$\mathcal{N}(a_n) = 1, \quad \mathcal{N}(a_n - u) = 1, \quad \mathcal{N}(2u - a_n) = 1 + \frac{2}{n}. \quad (2)$$

For the unit ball of any normed space, $\text{dist}(z, B) = \max\{\|z\| - 1, 0\}$; hence $\text{dist}(2u, B_E) = 1$. Furthermore

$$P_{B_E}(2u) = \{u\}. \quad (3)$$

Indeed, if $a \in B_E$ and $\mathcal{N}(2u - a) \leq 1$, then both a and $2u - a$ belong to B_E and have midpoint u . Extremality gives $a = u$.

Now fix $R = 1$ and $\varepsilon = 1/2$. For every $\delta > 0$, choose n so large that $2/n < \delta$. By (2), $a_n \in B_E$ and

$$\mathcal{N}(2u - a_n) < R + \delta.$$

Any $a' \in B_E$ satisfying $\mathcal{N}(2u - a') \leq R$ must equal u by (3), but then $\mathcal{N}(a_n - a') = 1 > \varepsilon$. This is the negation of uniform proximality for B_E . $\square$

4 The proper codimension-one counterexample

Theorem 1. *There are a real Banach space X and a proper closed codimension-one subspace $Y \subset X$ such that Y is uniformly proximal in X , with the modulus $\delta(\varepsilon, R) = \varepsilon$, but B_Y is not uniformly proximal in X . Consequently Problem 4.7 of [1] has a negative answer even under the additional requirement that Y be proper and codimension one.*

Proof. Let E be the Banach space of Lemma 1, and set

$$X = E \oplus_{\infty} \mathbb{R}, \quad Y = E \times \{0\}.$$

Plainly Y is closed, proper, and of codimension one. We prove uniform proximality with $\delta(\varepsilon, R) = \varepsilon$.

Take $R, \varepsilon > 0$, $z = (a, t) \in X$ with

$$\text{dist}(z, Y) = |t| \leq R,$$

and $y = (b, 0) \in Y$ with

$$\|z - y\|_X = \max\{\mathcal{N}(a - b), |t|\} < R + \varepsilon.$$

If $\mathcal{N}(a - b) \leq R$, take $y' = y$. Otherwise define

$$b' = a + \frac{R}{\mathcal{N}(b - a)}(b - a), \quad y' = (b', 0).$$

Then

$$\|z - y'\|_X = \max\{R, |t|\} = R$$

and, by homogeneity along the segment,

$$\|y - y'\|_X = \mathcal{N}(b - b') = \mathcal{N}(b - a) - R < \varepsilon.$$

Thus Y is uniformly proximal.

It remains to show that $B_Y = B_E \times \{0\}$ is not. Embed the obstruction from Lemma 1 by putting

$$z_0 = (2u, 0), \quad y_n = (a_n, 0).$$

The ℓ_{∞} norm gives

$$\text{dist}_X(z_0, B_Y) = \text{dist}_E(2u, B_E) = 1, \quad \|z_0 - y_n\|_X = 1 + \frac{2}{n}, \quad P_{B_Y}(z_0) = \{(u, 0)\}.$$

Also $\|y_n - (u, 0)\|_X = 1$. With $R = 1$ and $\varepsilon = 1/2$, choosing n so that $2/n < \delta$ defeats every proposed $\delta > 0$, exactly as in Lemma 1. Hence B_Y is not uniformly proximal. $\square$

5 Scope, novelty, and review recommendation

The construction is explicit and uses no computational or external mathematical dependency. The four lightweight run indexes and the packet and attempt indexes were searched for arXiv:1610.03209, “uniformly proximal”, “unit ball”, and “codimension one”. Bounded arXiv/web searches through 13 August 2026 used the exact wording of Problem 4.7 and combinations with “counterexample”, “proper subspace”, and “codimension-one”. They found the source paper and background literature on unit balls that fail strong or uniform proximality, but no answer to Problem 4.7 and no matching codimension-one construction.

There is an important scope distinction. The literal question allows $Y = X$, and the source’s own Remark 4.8(a) already makes that special case immediate. The potentially intended proper-subspace version is what Theorem 1 settles. Human review should verify the elementary extremality calculation, the exact norm values in (2), and the distance-preserving transfer to $E \oplus_{\infty} \mathbb{R}$. If these checks pass, the result is a complete negative answer to the strongest natural reading of the source question.

References

- [1] T. Paul, *Various notions of best approximation property in spaces of Bochner integrable functions*, Advances in Operator Theory **2** (2017), 59–77; arXiv:1610.03209v4.